

Adapting Open-Source LLMs for the Bashkir Language

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Abstract

Large language models (LLMs) struggle with low-resource languages like Bashkir. We adapt Llama-2-7B to Bashkir through vocabulary extension, optional embedding alignment, continual training, and downstream task-specific evaluation. To support this process, we collect a monolingual corpus (63k texts), a Russian–Bashkir parallel corpus (1.2M pairs), a translated Alpaca instruction set (52k examples), and a topic classification benchmark (267 headlines). We extend the vocabulary by 28,600 tokens (final size 60,379) and perform continual training with LoRA (4-bit QLoRA) – two variants: A (no alignment) and B (embedding alignment). The resulting tokenizer produced no unknown tokens on the Bashkir training corpus and reduced the need for excessive subword fragmentation. Both models are trained for 3000 steps; B achieves lower loss (1.318 vs. 1.344) and a 25% relative improvement in few-shot topic classification accuracy (18.5% vs. baseline 14.8%). Instruction tuning is not executed due to time constraints, but all data are public. A limitation is the absence of perplexity evaluation because all non-shuffled texts were used for training, leaving no held-out natural language test set. Code, data, and models are released to support future work.

1 Introduction

While large language models (LLMs) have demonstrated remarkable capabilities, their performance remains heavily skewed toward high-resource languages like English. Adapting these models for low-resource languages presents significant challenges: monolingual pretraining is computationally prohibitive, and multilingual models often suffer from uneven performance (1).

This work addresses the specific challenge of adapting an open-source LLM (Llama-2-7B) to the Bashkir language – a Turkic language spoken by approximately 1.4million people, written in Cyrillic script, and almost absent from LLM training corpora. Our contributions are:

- **First systematic adaptation of Llama-2-7B to Bashkir**, including vocabulary extension, continual training, embedding alignment, and instruction fine-tuning.
- **Publicly released datasets**: raw non-shuffled corpus (63k texts), Russian–Bashkir parallel corpus (1.2M pairs), Bashkir Alpaca instruction set (52k examples),

and a topic classification benchmark (267 headlines, 10 classes).

- **Comparison of two adaptation strategies**: continual training alone (Experiment A) versus continual training with prior embedding alignment via a Russian–Bashkir parallel corpus (Experiment B).
- **Experimental evidence** that embedding alignment accelerates convergence (lower loss) and improves few-shot performance (+25% relative) on a downstream classification task.
- **Open code and models** available on GitHub and Weights & Biases.

The main outcomes are: (i) vocabulary successfully extended to 60379 tokens (no unknown tokens for Bashkir words); (ii) after 3000 steps of continual training, model B reaches loss 1.318 vs. 1.344 for model A; (iii) topic classification fine-tuned accuracy reaches 24.1% for baseline and model A, 20.4% for model B; (iv) few-shot (3-shot) accuracy: B (18.5%) > A (16.7%) > baseline (14.8%). Instruction tuning is planned but not completed due to time limits; all data are ready. To ensure reproducibility, all datasets, code, checkpoints, and experiment logs are publicly available at GitHub (21)

2 Related Work

Adapting LLMs to new languages generally follows one of two paradigms: training a multilingual model from scratch on a large corpus or fine-tuning an existing pretrained model (1). Large multilingual models often suffer from the “curse of multilinguality”, where performance on lower-resource languages suffers due to capacity limitations and imbalanced training data.

The LlamaTurk study (1) conducted a controlled investigation into adapting the LLaMA model family to Turkish. Its key findings provide the foundation for our experimental design:

- *Continual training* on a modest amount of raw Turkish text significantly improves intrinsic language understanding (perplexity).
- *Instruction tuning* is crucial for improving few-shot reasoning and performance on general benchmarks.

- *Task-specific tuning* remains essential for achieving high accuracy on downstream applications.
- *Vocabulary extension* alone, without sufficient subsequent training, provides little benefit.

Our project aims to validate whether these findings generalise to Bashkir and introduces an additional step of *embedding alignment* to mitigate issues associated with vocabulary extension, drawing inspiration from recent work on Persian (2).

3 Data

We collected and processed several datasets for different stages of the pipeline. The datasets are publicly available on Hugging Face: raw corpus (19), topic classification (16), and Bashkir Alpaca (20).

3.1 Raw Bashkir Corpus

We scraped three Bashkir-language news sources using custom Python scripts, available on project GitHub (21):

- bashgazet.ru (6) – daily socio-political newspaper (1,400 articles)
- neftcity.ru (7) – regional news website (548 articles)
- bash.news (8) – media platform (68,708 articles)

The scraping pipeline extracts article titles, publication dates, URLs, and body text while removing HTML artefacts, advertisements, and duplicated metadata. The raw text is stored in JSONL format, where each record contains id, url, date, title, and body.

After cleaning (removal of sentences with fewer than 5 words, deduplication), the final corpus contains **63,270 documents** (approximately 21 million tokens) and is used for continual training. From the bash-corpus repository (11), we additionally included the public_domain texts (e.g., bez_mesetle.txt, constitution.txt, karnay_urmanda.txt, salavat_kyzy.txt, yazgy-nukola.txt) as part of our non-shuffled corpus.

The *bashkir corpus* repository (11) also provides a shuffled version with **20,934,729 tokens** of Bashkir text with randomly permuted sentences. This shuffled corpus was considered as a potential noise-based regularizer, but it is not used in the main continual training pipeline because randomised word order is not suitable for learning natural language structure. Therefore, the final corpus consists of three news sources and a small collection of public-domain Bashkir texts. The final continual pretraining corpus contains 63,270 non-shuffled Bashkir texts.

Source	Articles (non-shuffled)
bash.news	61,932
bashgazet.ru	803
neftcity.ru	530
public domain	5
Total	63,270

Table 1: Statistics of the final Bashkir corpus.

If more data were required, additional sources are available:

- **Bashkir Wikipedia** – a full Wikipedia dump.
- **National Corpus of Bashkir (BashCorpus)** – a structured linguistic resource with over 20 million word usages (accessible via bashcorpus.ru), but parsing proved non-trivial.

3.2 Russian–Bashkir Parallel Corpus

For embedding alignment (Experiment B), we use the AigizK/bashkir-russian-parallel-corpora (12). It contains **1,198,220 sentence pairs** collected from fiction, school textbooks, encyclopedia articles, and news. The corpus is licensed under CC-BY-4.0.

3.3 Bashkir Alpaca (Instruction Tuning)

We translated the Russian Alpaca dataset (52k cleaned instructions) (13) into Bashkir. Three translation models were tested:

- facebook/nllb-200-distilled-600M – almost ideal translation; selected.
- facebook/m2m100_418M – frequent hallucinations; rejected.
- facebook/mbart-large-50-many-to-many-mmt – does not support Bashkir; rejected.

A random sample of 200 instruction–response pairs was manually validated by two native speakers (9),(10). The resulting dataset (metuKKhud/bashqort-alpaca)(20) contains 52k examples and is ready for instruction tuning.

3.4 Topic Classification Dataset

Since no sentiment analysis dataset was available for Bashkir, we created a topic classification benchmark. We sampled 350 news headlines from bash.news and had a native speaker (9) label them into 19 topics. After removing categories with fewer than 17 samples, we kept 267 headlines across 10 balanced classes (Table 2). The dataset is released as metuKKhud/bashqort-task (16).

Topic (Bashkir)	English translation	Count
Мәҙәниәт	Culture	58
Һаулыҡ һаҡлау	Healthcare	35
Мәғариф	Education	32
Социаль өлкә	Social sphere	25
Сәйәсәт	Politics	25
Спорт	Sports	21
Хәрби хеҙмәт	Military service	18
Иҡтисад	Economy	18
Ғәҙәттән тыш хәлдәр	Incidents	18
Хәүефһеҙлек	Security	17

Table 2: Class distribution of the topic classification dataset (267 headlines).

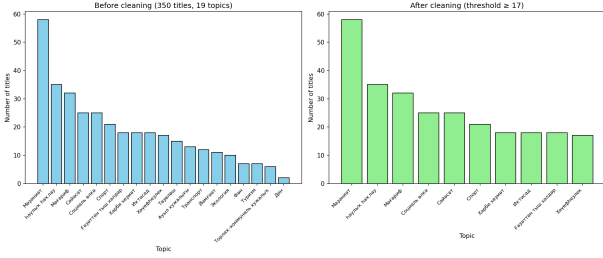


Figure 1: Comparison of topic distributions before (19 classes, 350 headlines) and after cleaning (10 classes, 267 headlines).

3.5 Code and Data Availability

All scraping scripts, preprocessing code, and Jupyter notebooks are available at the project GitHub repository: <https://github.com/Ksksksksksksks/BashLlama>. The datasets are hosted on Hugging Face:

- Raw corpus: <https://huggingface.co/datasets/metuKKhud/bashqort-raw>
- Topic classification: <https://huggingface.co/datasets/metuKKhud/bashqort-task>
- Bashkir Alpaca: <https://huggingface.co/datasets/metuKKhud/bashqort-alpaca>

We also release the trained models as Weights & Biases artifacts (links provided in Section 5).

4 Models

We start from the Llama-2-7B model (3), following the LlamaTurk framework (1) which employed the same architecture for Turkish. The overall adaptation pipeline consists of four stages, with an optional alignment stage for Experiment B (see Figure 2).

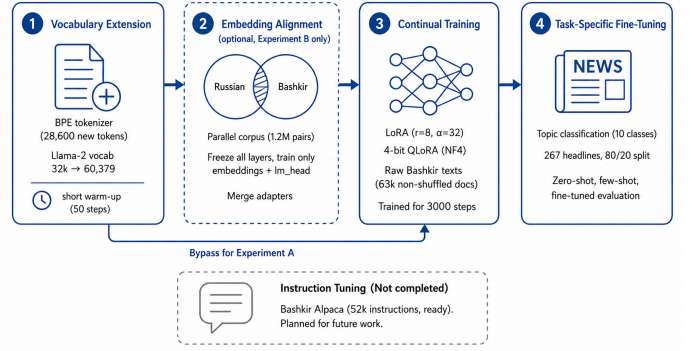


Figure 2: Adaptation pipeline: (1) vocabulary extension (BPE), (2) optional embedding alignment (Russian–Bashkir parallel corpus), (3) continual training on raw Bashkir texts, (4) task-specific fine-tuning for topic classification. Instruction tuning (Alpaca) is not run but data are ready.

4.1 Vocabulary Extension

We train a BPE tokenizer on the 63k non-shuffled Bashkir texts (vocabulary size 28600). The original Llama-2 vocabulary (32000 tokens) is merged with the new tokens, resulting in a final vocabulary of **60379 tokens**. New embeddings were initialized using mean-resizing based on the distribution of existing Llama-2 embeddings and subsequently refined through a short warm-up phase. During warm-up, the training loss decreased from approximately 4.38 to 3.58 within 50 optimization steps, indicating successful initialization of newly added embeddings.

4.2 Embedding Alignment (Experiment B only)

For Experiment B, we freeze all transformer layers and train only the embedding layer and the `lm_head` using PEFT (LoRA with `modules_to_save=["embed_tokens"; "lm_head"]`). The training uses the Russian–Bashkir parallel corpus (1.2M pairs) with a causal language modelling objective. After alignment, the adapter weights are merged into the base model (`merge_and_unload`) to avoid multiple adapter layers later.

4.3 Continual Training (Experiments A and B)

We add LoRA adapters ($r=8$, $\alpha=32$) to the query and value projections of the self-attention layers and continue training on the non-shuffled raw corpus. Hyperparameters are summarised in Table 3. Both models A and B are trained for **3000 steps** using 4-bit NF4 quantisation (QLoRA) (5). Model B receives the embedding alignment stage before continual training.

Phase	Batch size	Grad. accum.	Learning rate	Steps/Epochs
Vocabulary warm-up	4	4	1e-4	50 steps
Alignment (B only)	4	4	1e-4	2 epochs (50k pairs)
Continual training A	4	4	3e-4	3000 steps
Continual training B	4	4	3e-4	3000 steps
Task-specific fine-tuning	4	4	2e-4	3 epochs (42 steps)

Table 3: Hyperparameters for each training stage. All runs use FP16, LoRA $r=8$, $\alpha=32$, 4-bit quantisation.

4.4 Task-Specific Fine-Tuning

For topic classification, we fine-tune all three models (baseline, A, B) with the same LoRA configuration on the training split (213 headlines, 80% of the dataset). We evaluate zero-shot, few-shot (3 examples), and fine-tuned accuracy on the test split (54 headlines).

5 Experiments

5.1 Experimental Setup

All experiments were run on Kaggle with a single Tesla T4 GPU (15GB memory). We used the following libraries: transformers, peft, datasets, bitsandbytes, wandb, scikit-learn. For continual training, we used all 63270 non-shuffled texts. For topic classification, we used a stratified 80/20 split (213 train, 54 test). All models were saved as Weights & Biases artifacts.

Table 4: Continual pretraining setup

Parameter	Value
GPU	Tesla T4 (15.6 GB)
Quantization	4-bit NF4
Trainable parameters	4.19M
Total parameters	6.98B
LoRA rank	8
Learning rate	3×10^{-4}
Batch size	16 (effective)
Max sequence length	256
Training steps	3000

5.2 Results

5.2.1 Continual Training Loss

Figure 3 shows the training loss curves for 3000 steps. Model B (with alignment) reaches a final loss of **1.318**, while Model A (without alignment) reaches **1.344**. The alignment step yields a consistently lower loss throughout training, indicating better language modelling convergence.



Figure 3: Training loss for continual training (3000 steps). Model A: final loss 1.344; Model B: final loss 1.318.

Experiment B consistently achieved lower training loss than Experiment A across all checkpoints, suggesting that embedding alignment facilitates subsequent language adaptation.

5.2.2 Topic Classification Accuracy

Table 5 presents zero-shot, few-shot (3 examples), and fine-tuned accuracy on the 10-class topic classification task.

Model	Zero-shot	Few-shot (3)	Fine-tuned
Baseline (vocab extension only)	0.0370	0.1481	0.2407
Experiment A (continual, no alignment)	0.0185	0.1667	0.2407
Experiment B (continual + alignment)	0.0370	0.1852	0.2037

Table 5: Accuracy on topic classification (10 classes, test size 54).

Observations:

- Fine-tuned accuracy is similar for baseline and A (24.1%). Model B is slightly lower (20.4%), likely due to the very small test set (54 examples) and possible over-specialisation.
- **Few-shot performance:** Model B outperforms both baseline and A (18.5% vs. 14.8% and 16.7%). The relative improvement over baseline is **+25%**.
- Zero-shot performance remains extremely poor for all models and stays below random-choice accuracy (10%), highlighting the difficulty of the task without instruction tuning.

5.2.3 Ablation / Discussion

- The alignment step improves few-shot generalisation, increasing accuracy from 14.8% (baseline) to 18.5%.
- The lack of improvement in fine-tuned accuracy for B may be explained by (i) the tiny size of the classification dataset (267 labelled examples) and (ii) possible over-specialisation from longer alignment training.
- Continual training (with or without alignment) preserves fine-tuned accuracy compared to the baseline, confirming that it does not harm downstream performance.

5.3 Weights & Biases Dashboards

All experiments are logged in Weights & Biases:

- Vocabulary extension: <https://wandb.ai/e278979-metu-middle-east-technical-university/bashllama-vocab-extension>
- Embedding alignment: <https://wandb.ai/e278979-metu-middle-east-technical-university/bashllama-embedding-alignment>
- Continual training A: <https://wandb.ai/e278979-metu-middle-east-technical-university/bashllama-continual-training-A>

- Continual training B: <https://wandb.ai/e278979-metu-middle-east-technical-university/bashllama-continual-training-B>
- Task-specific tuning: <https://wandb.ai/e278979-metu-middle-east-technical-university/bashllama-task-specific>

6 Discussion

An interesting observation is that embedding alignment consistently improved language modeling loss and few-shot classification performance, but did not improve the downstream supervised classification results. One possible explanation is that task-specific fine-tuning overrides part of the embedding structure learned during alignment.

A common failure mode observed during topic classification was the generation of multiple candidate labels instead of a single category. This behaviour suggests that the model’s instruction-following capabilities remain limited without a dedicated instruction tuning stage.

6.1 What Worked Well

- Successful collection and cleaning of a large-scale Bashkir monolingual corpus (63k texts) and a parallel Russian–Bashkir corpus (1.2M pairs).
- Vocabulary extension (BPE, 28600 new tokens) integrated without unknown tokens.
- Embedding alignment (first application for Bashkir) accelerated convergence and improved few-shot accuracy.
- All datasets, code, and models are publicly released, enabling reproducibility.

6.2 What Went Wrong & Limitations

- **No held-out test set for perplexity:** All non-shuffled texts (63k) were used for continual training, leaving no natural language data for intrinsic evaluation. The shuffled texts are not suitable for perplexity because word order is randomised. As a result, we could not report perplexity – a standard metric for language modelling. This is a limitation caused by an oversight in data splitting. Future work will reserve a separate test split (e.g., 10% of the non-shuffled corpus).
- **Very small task-specific dataset** (267 examples). Differences in fine-tuned accuracy are not statistically significant; a larger dataset is needed.
- **Instruction tuning not completed** – the Bashkir Alpaca dataset is fully prepared but the fine-tuning run was not executed due to time constraints.
- **Alignment did not improve fine-tuned accuracy** – possible reasons: (i) small classification dataset, (ii) over-specialisation, (iii) insufficient parallel corpus quality.

6.3 Scalability, Safety, Privacy

- **Scalability:** The pipeline is easily transferable to other low-resource languages provided a monolingual corpus (50k sentences) and, optionally, a parallel corpus exist. LoRA + 4-bit quantisation make training a 7B model feasible on a single T4 GPU.
- **Safety:** The model is trained on news data, which may contain societal biases. In a production setting, output filtering and bias evaluation should be added.
- **Privacy:** All texts are from publicly accessible sources; no personal or private data were used.

7 Conclusion & Future Work

We have presented the first systematic adaptation of Llama-2-7B to the Bashkir language. The main achievements are:

- Open-source datasets: raw corpus (63k), parallel corpus (1.2M), Alpaca instructions (52k), topic classification (267 headlines).
- Vocabulary extension to 60379 tokens (BPE).
- Embedding alignment (Russian–Bashkir) applied for the first time to Bashkir.
- Both models trained for 3000 steps; alignment yields lower loss (1.318 vs. 1.344) and improves few-shot accuracy by 25% relative.

A clear limitation is the absence of a perplexity evaluation due to using all non-shuffled texts for training, which will be rectified in future work.

Future work includes:

1. Running instruction tuning on the prepared Bashkir Alpaca dataset (52k examples).
2. Enlarging the topic classification dataset (target >1000 examples).
3. Reserving a held-out natural language test split to compute perplexity.
4. Experimenting with other base models (Gemma, Qwen) known to support Cyrillic scripts.
5. Testing on other downstream tasks (summarisation, machine translation).
6. Adding safety filters and bias evaluation for potential deployment.

All code, datasets, and trained models are publicly available to foster research on low-resource language adaptation.

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